interval to 17.2 s after a 8,192 min (5.67 days) interval. The median avoidance scores (Fig. 1) revealed a biphasic time course of retention that suggests more than one process is active in their memory formation.

The avoidance data were analysed in overall and individual comparisons, using the Kruskal-Wallis one-day analysis by ranks, the chi-square test, and the Mann-Whitney U test. At the 0.5 min interval, trained fish performed as they had before exposure to the acid; this may indicate an acid-induced alarm reaction or a failure to organize the pertinent information to avoid. During the overall period of 1–4,096 min after training, acid-trained fish avoided significantly longer than did the corresponding controls (P, 0.01). The trained behaviour was time-dependent; at 4 min, the avoidance was less than at 2 min or 8 min (P, 0.02 and 0.009), although it did not fall to the control or pretraining levels (P, 0.03 and 0.001, respectively). That this time-dependent dip was not the result of random fluctuations was shown by tests of the proportion of fish that avoided for more than 30 s. In a 2×10 contingency table for the ten intervals from 1 to 128 min, $\chi^2 = 17.30$, df 9, $P < 0.05$; for the five intervals from 1 to 16 min, $\chi^2 = 8.43$, df 4, $P < 0.08$. At the 64 min interval, an apparent second dip prompted us to test retention of avoidance in additional groups at the 45 min and 99 min intervals. Rank analyses of these data led us to conclude that the inflexion at 64 min resulted from influences not directly related to memory. At the two end points of curve (0.5 min and 8,192 min), the median avoidances did not exceed the control level ($P = 0.14$).

The avoidance of acid-trained fish varied considerably within a test interval group but their scores met the Cochran C test for homogeneity of variance ($C = 0.1039$, $P > 0.05$). The surprisingly high dispersion of scores was also reflected in the high but uniform semi-interquartile ranges at each interval (26.0–46.2 s), inconveniently large for graphical display. This dispersion is ascribed to phenotypic variations because genetic differences can be discounted in unisexual fishes; nevertheless, we tested the degree of relation among the scores of different spawns and found that the biserial correlation ranged from $r_b = 0.62$ to $+0.88$.

Influences on the shape of the retention curve, other than time-dependent retrieval of memory, cannot be ruled out. As the testing schedule placed the fish under observation at different times of the 12 h light cycle, changes in daily activity pattern could have contributed to the dispersed scores within a test interval; controls, however, underwent the same regimen and their scores did not deviate noticeably from their pretraining scores, even when tested repeatedly. Neither can the apparently anomalous dip in retention be attributed to non-specific after-effects of the acid treatment, since trained fish and non-contingent controls had shared the same exposure to acid.

Dips in retention performance at 3 min in chicks[12], at 5 min in mice[13], at 22 min in *Sepia*[14], at 30 min in mice[15–16], at 1 h in rats[17] and in the headless cockroach preparation[18], and at 1 and 8 h in *Octopus*[19] have been reported. We have previously found that goldfish show an apparent amnesia at 1 min after one-trial avoidance training and that this dip is dependent upon temperature and upon the intensity of the aversive shock[20–21]. Most of these curves differ from each other in the interval at which the retention minimum is found. The results have been observed within a limited set of training conditions and generally under "weak" stimulation. We show the existence of a distinct early phase with a brief and sharp rise in retention performance and also that the extent of avoidance is time dependent.

The apparent retention deficit has divergent interpretations. Some hold the accumulation of stress-related physiological changes responsible for the retrieval failure[22], others motivational deficits[23], and others the incubation of "fear", that is, the gradual increase and later dissipation of responses incompatible with the avoidance activity[24]. These interpretations assume that additional influences interfere normally with memory processing. Our results may demonstrate two phases in memory formation, each with different rise and fall times, as postulated by biphasic models of memory processing[25]. The transition phases would become observable from such a temporary minimum in retention. The course of retention in a unisexual fish may thus help to delineate how memory is processed.

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Membrane Characteristics of Bursting Pacemaker Neurones in Aplysia

THE group of endogenously active molluscan neurones, classified as bursting or oscillatory pacemakers, regularly fire bursts of three to fifteen impulses separated by periods of silence associated with membrane hyperpolarizations (Fig. 1*a*). The abdominal ganglion of the marine mollusc *Aplysia californica* contains six bursting pacemakers, L2–L6 (L-group) and R15[1]; these cells maintain their membrane potential oscillations in the absence of synaptic input[2,3] and spike generation[4]. They also have K^+-dependent inhibitory postsynaptic potentials (IPSPs) which last for some seconds after single or multiple discharges in the presynaptic neurones[5–8]. In the L-group these IPSPs are mixed with short-lasting Cl^--dependent IPSPs. Although there is a different transmitter responsible for the generation of these IPSPs in R15 as opposed to those

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in the L-group, somewhat similar difficulties were encountered in attempts to determine the mechanism(s) underlying their generation, activation of an electrogenic Na^+ pump having been initially implicated in both cases[5,8]. Just why their duration is prolonged remains uncertain, but the voltage clamp studies of Wachtel and Wilson[9] on the L-group have again implicated a regenerative current source.

We have investigated the membrane characteristics of these neurones using standard techniques of intracellular voltage recording and voltage clamping. A description of their behaviour requires the consideration of at least two properties not found in classical axon or neurone models.

The progressive changes occurring within a burst[10] are as follows (Fig. 1*a* and *b*): (i) an increase in spike overshoot, (ii) a simultaneous increase in spike duration and enhancement of the shoulder on the falling phase of the spike, (iii) a decrease in the hyperpolarizing afterpotential, and (iv) an initial decrease in the interspike interval followed by an increase (that is, the interspike interval is a parabolic function of time). The last characteristic is best seen in R15 which generally has the most spikes per burst. Also, the post-burst hyperpolarization does not decay exponentially but sigmoidally.

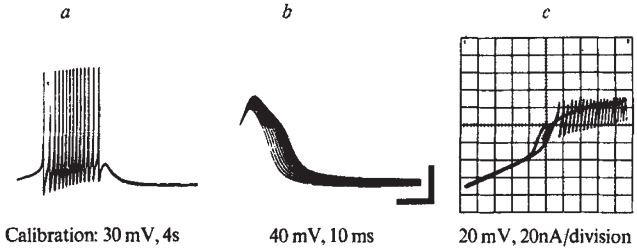


Fig. 1 *a*, Increase in spike overshoot and decrease of the hyperpolarizing afterpotential in cell L-2 during a burst following a hyperpolarization. *b*, Changes of the action potential during a single burst in cell L-2. The first spike is the lower one, having smaller overshoot and shorter duration. During the train, overshoot increases by about 12 mV. *c*, Current-voltage relationship during current clamp of cell L-4. Pronounced anomalous rectification around 20–30 mV above resting potential. Current calibration refers to horizontal axis; current changed by 3 nA/s.

The I–V relationship of these cells, determined by applying linearly increasing and decreasing de- and hyperpolarizing transmembrane currents with one intracellular electrode and measuring membrane potential with a second, exhibits pronounced anomalous rectification (Fig. 1*c*). In other words, the membrane resistance decreases when the cells are hyperpolarized. The importance of anomalous rectification for bursting is shown by our observation that when the convulsant pentylenetetrazol causes similar bursting in normally silent cells, it either introduces or enhances anomalous rectification (unpublished work).

The characteristic that distinguishes these cells in voltage clamp from the others in the ganglion is a subthreshold fast outward current, I_A, presumably carried by potassium ions (Fig. 2*a*). First described by Stevens for *Anisodoris* neurones[11] and by Neher and Lux for *Helix* neurones[12], I_A is activated by depolarizing command pulses only when the initial membrane potential is hyperpolarized by at least 10 mV beyond the resting potential. Fig. 2*a* illustrates the early transient currents evoked by a test depolarization of 40 mV when it is preceded by 400 ms conditioning hyperpolarizations of 10 to 40 mV, and when there is no conditioning pulse. The figure shows that in *Aplysia* the activation of I_A has a time course comparable to the inward current and masks the latter. While previous descriptions of I_A did not indicate the behaviour of the cells in which it was found, we found that it is only clearly observable in *Aplysia* in the bursting pacemakers. In the other cells, a distinct fast outward current is never seen with our technique, and its manifestation is at best a 10–35% reduction in the inward current after conditioning hyperpolarizing pulses.

I_A is assumed to take part in prolonging interspike intervals, being activated during the recovery of the membrane potential from hyperpolarizations such as hyperpolarizing spike afterpotentials or IPSPs. In the bursting pacemakers it would be activated by the decay of a post-burst hyperpolarization and would then be inactivated by the maintained depolarization (Fig. 2*b*), with at best partial reactivation occurring after each spike. As such, it explains at least two characteristics of a burst. First, it would make the initial interspike intervals long, the intervals shortening as I_A inactivated. Second, it accounts for the increase in spike overshoot during a burst. Simultaneous activation of the opposing fast outward and inward currents would give a smaller spike overshoot at the beginning of a burst than at a later time when I_A is no longer activated. The analogue to this in voltage clamp is a shift to lower voltages of the equilibrium potential of the "apparent" inward current when I_A is also activated. Such a shift was predicted by Neher[13] and we have confirmed his prediction, finding shifts of the order of 10–20 mV when the holding potential is changed from −40 to −70 mV.

The other characteristics of a burst which can partly be attributed to the initial activation of I_A with each spike and its subsequent inactivation are the progressive increase in spike duration and the reduction in afterpotential. However, we feel that these properties are also due to the inactivation of the classical delayed rectification potassium current (Fig. 2*c*) originally described by Frankenhauser and Hodgkin[14] for the squid giant axon. In that preparation K^+-inactivation accounts for the decreases in afterpotential and overshoot during a train of action potentials.

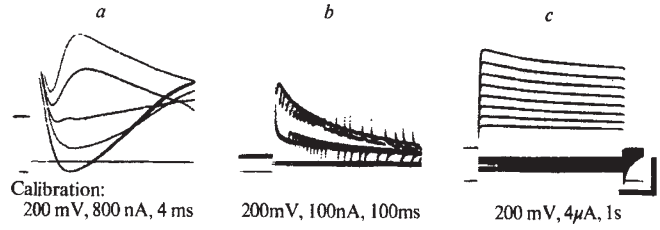


Fig. 2 *a*, Superposition of the early inward (downward) and the fast outward currents of cell L-4. From a holding potential (*Vh*) of −40 mV, the cell was depolarized by a command (*Vc*) to zero voltage, producing the record with the maximum inward current. Following 400 ms hyperpolarizing prepulses of −15, −20, −25 and −40 mV, the same command activated the early outward current system as well. *b*, Time course of inactivation of the early outward current in cell R-15. From a hyperpolarized *Vh* of −80 mV, the membrane potential was first reduced to −40 mV, the resting potential of the cell, and after progressively increasing intervals further reduced to −27 mV for 40 ms. The latter potential shift produced the early outward current, the amplitude of which decayed exponentially as a function of the duration of the intermediate step. Superimposed is the outward current and its inactivation time course when the potential was stepped directly to −27 mV and maintained at that level for 700 ms. Its time course is virtually the same as the envelope of the peak outward currents. *c*, Inactivation of the delayed rectification potassium current during depolarization of cell L-6 for 4 s. *Vh* = −40 mV, *Vc* = +10, +20 to +80 mV.

We suggest that both I_A and anomalous rectification are important for the membrane potential trajectory between bursts and for the prolonged duration of the long-lasting IPSPs. As the current sources underlying these hyperpolarizations decay both mechanisms will counteract the return of the membrane potential to its resting level. The transition to the high resistance region will account for the rapid decay phase of these hyperpolarizations. Finally, similarities of ionic and temperature dependency between anomalous rectification and I_A[15,16] lead us to wonder to what extent they, and possibly

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also K^+-inactivation and the current system described by Wachtel and Wilson[9], reflect linked processes.

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Errata

THE illustrations and legends in the article "Preferential Sensitization of Hypoxic Cells to Radiation *in vivo*" by J. Denekamp and B. D. Michael (*Nature New Biology*, **239**, 21; 1972) were unfortunately incorrect. The correct versions of Figs. 1 and 2 appear here.

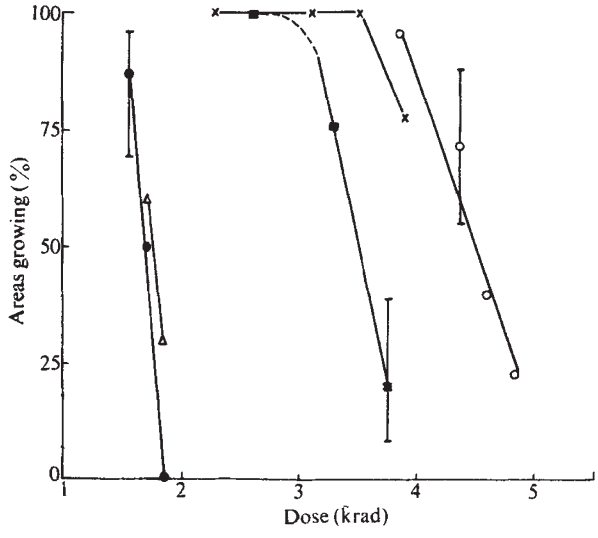


Fig. 1 Survival of epidermal clones as a function of the test dose administered to animals breathing oxygen or nitrogen, with or without intravenous administration of NDPP (CRC 55, Ro 3–6156). The vertical bars are 95% confidence limits based on the sample size.

●, Oxygen; △, 1 mg NDPP+oxygen; ×, 1 mg NDPP+nitrogen; □, 3 mg NDPP+nitrogen; ○, nitrogen.

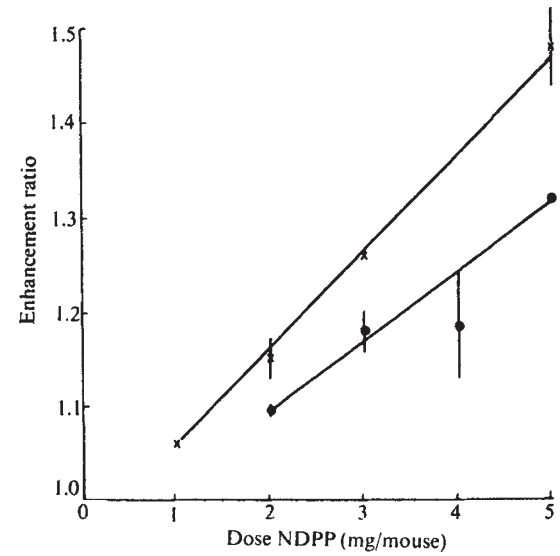


Fig. 2 Enhancement ratio for the survival of epidermal clones in nitrogen-breathing animals as a function of the dose of NDPP (CRC 55, Ro 3–6156) administered intravenously. ×, WH mice; ●, C_3H mice.

IN the article "LRF and CRF Release in Subcellular Fractions of Bovine Median Eminence" by G. Fink, G. C. Smith, J. Tibballs and V. W. K. Lee (*Nature New Biology*, **239**, 57; 1972), the legend to Fig. 3 should read: Electron micrograph of fraction P3D showing a structure (S) which contains numerous agranular vesicles. This structure is presumed to be a disrupted nerve terminal. Profiles resembling granular vesicles are seen nearby (arrow).

THE following corrections should be made to the article "Homology between RNA from Human Malignant Breast Tissue and DNA synthesized by Milk Particles" by M. R. Das, E. Sadasivan, R. Koshy, A. B. Vaidya and Satyavati M. Sirsat (*Nature New Biology*, **239**, 92; 1972): on page 92, paragraph 2, line 1, Parsis should read Parsi; in the legend to Fig. 1, line 1, 250 ml. should read 250 µl.; in the legend to Table 1, line 4, 50 aliquots should read 50 µl. aliquots; in the legend to Table 2, line 1, 500–600 g should read 500–600 µg; and on page 94, line 8 of column 2 should read "RNA AC 12928 and DNA product from milk sample H9".

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